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SUPERSEDING
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MILITARY SPECIFICATION

PANELBOARDS, POWER DISTRIBUTION, PORTABLE, WEATHERPROOF, GENERAL SPECIFICATION FOR

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric power distribution panelboards, hereinafter referred to as distribution panels, mounted in a portable, weatherproof housing.

1.2 Classification. The distribution panels will be of the following types, classes, and configuration, as specified (see 6.2):

Type I - Transformerless type distribution panels.
Type II - Transformer type distribution panels.

Class 1 - 3 Phase 60 Hertz (Hz) input power.
Class 2 - 1 Phase 60 Hz input power.

Configuration ( ) - As delineated in the applicable specification sheet title.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6110

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

W-C-375 - Circuit Breakers, Molded Case; Branch Circuit and Service (General Specification).
W-C-596 - Connector, Electrical Power, General Specification for.
TT-E-1593 - Enamel, Silicone Alkyd Copolymer, Gloss (For Exterior and Interior Use).

MILITARY

MIL-T-152 - Treatment, Moisture-and-Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-T-704 - Treatment and Painting of Material
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.
MIL-C-22992 - Connector, Plugs, and Receptacles, Electrical, Waterproof, Quick Disconnect, Heavy Duty Type, General Specifications for.
MIL-C-46168 - Coating, Aliphatic Polyurethane, Chemical Agent Resistant

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-595 - Colors.

MILITARY

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-889 - Dissimilar Metals.

MS 18012 - Handle, metal, spring bail type.

DOD-STD-2000-4 General purpose soldering requirement for electrical and electronic equipment.

(See Supplement 1 for list of associated specifications).

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(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Military Specification and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which is current on the date of the solicitation (see 6.2).

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA WD 1 - General requirements for wiring devices.
NEMA WD 6 - Wiring devices - dimensional requirements.
NEMA ST 20 - Dry type transformers for general applications.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

UNDERWRITERS LABORATORIES INC. (UL)

UL 67 - Electric Panelboards.
UL 486A - Wire Connectors and Soldering Lugs for Use with Copper Connectors.
UL 489 - Molded-Case Circuit Breakers and Circuit-Breaker Enclosures.
UL 506 - Speciality Transformers.
UL 943 - Ground Fault Circuit Interrupters.

(Application for copies should be addressed to the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.1.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern (see 6.2).

3.2 Description. The distribution panels shall be factory assembled, packaged, load center type with low profile, low center of gravity, arranged in a horizontal configuration and contained in a rainproof and sleet (ice) resistant ventilated enclosure.

3.3 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.1).

3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Unless otherwise specified, none of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification.

3.5 Identical items. All distribution panels of the same type as specified in the applicable specification sheet, furnished under any specific contract shall be physically and mechanically identical. This requirement includes parts, assemblies, components and accessories. No changes or deviations will be acceptable without written approval of the contracting officer.

3.6 Codes and standards. The panels and components shall conform to the requirements of NEMA ST 20, NEMA WD 1, NEMA WD 6, UL 67, UL 486A, UL 489, UL 506, UL 943 and NFPA 70 as applicable.

3.6.1 Compliance. Prior to approval of the first shipment, the contractor shall submit to the contracting officer, or the authorized representative, satisfactory evidence that the panelboard and components contractor proposes to furnish under this specification meets the requirements of NEMA ST 20, NEMA WD 1, WD 6, UL 67, UL 486A, UL 489, UL 506, UL 943 and NFPA 70 as applicable.

3.6.2 Underwriters Laboratories. Acceptable evidence of meeting the requirements of UL 67 shall be the UL certification symbol or label, listing in the listing files or a certified test report from a recognized independent testing laboratory indicating the panelboard has been tested and conforms to UL 67. Such evidence must be acceptable to the contracting officer.

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3.7 Specific design. Unless otherwise specified in the applicable specification sheet, the distribution panel design shall include the following specific design features:

a. The distribution panels shall be mounted on a skid-type frame base.
b. Rigid base end frames with "A" frame type construction.
c. Rigid longitudinal horizontal members connecting the two end frames. The horizontal members shall be suitable for the lifting of the distribution panel by forklift.
d. Inward sloping connector panel faces for weather protection of connector outlets and ease of cable insertion and removal.
e. Low base mount profile for stability.

3.8 General design. The distribution panels shall be designed to distribute electrical power from commercial power or as applicable, engine-generator sets, operating individually or in parallel and shall be in accordance with the applicable specification sheet as specified in the acquisition document. Each distribution panel shall be contained in a single assembly and have the capability of operating individually or being connected for parallel operation with other similar distribution panels. The distribution panel design and component installation shall permit accessibility for maintenance and service in the field. The design shall be such as to prevent conditions which may be hazardous to personnel or deleterious to equipment. The general panel assembly design shall be in accordance with UL 67 and NFPA 70.

3.8.1 Metals. Metals, other than current-carrying parts, shall be the corrosion-resistant type, unless suitably protected to resist corrosion during the normal service life of the equipment.

3.8.1.1 Dissimilar metals. Dissimilar metal contact to the degree defined in MIL-STD-889 shall be avoided. When such contact cannot be avoided, an interposing insulating material shall be installed in accordance with MIL-STD-889.

3.8.1.2 Protective treatment. The distribution panel shall be protected against corrosion from salt laden or other atmosphere conditions in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or the extremes of atmosphere conditions shall be avoided. A suitable passivation treatment shall be accomplished after fabrication and assembly of all parts of the distribution panel and the skid frame base that could be exposed and subjected to the above extreme corrosion conditions.

3.8.2 Hardware. The different sizes and types of hardware shall be kept to a minimum. All nuts, bolts, screws, and threaded parts shall have standard screw threads in accordance with FED-STD-H28. Fasteners with metric threads and heads shall not be used. All exterior shall threaded parts shall be made of stainless steel or plated brass. Sheet metal screws shall not be used.

3.8.3 Construction and stability. The distribution panel shall be constructed to withstand the strains, vibrations, rough handling, and other conditions incident to shipping, storage, and service at military

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installations. The distribution panels shall be fabricated from lightweight, heavy-gage, high-strength aluminum, constructed with suitable integral structural members so that the components form a compact and stable unit, and so that adjustment, repairs, and replacement of parts can be easily made by operating military personnel in the field. All terminals for exterior connections and manual controls shall be conveniently located. The distribution panel shall be stable when the skid-frame base is located on uneven surfaces or on slopes of as much as 15 degrees (^{o}) from the horizontal in any direction (see 4.4.5).

3.8.4 Welding. Portions of the distribution panel enclosure and of the skid-frame base shall be fabricated by welding rather than by bolting or riveting. Surfaces to be welded shall be free from foreign matter which would be injurious to the weld. Welding procedures shall be in accordance with a nationally recognized code. Welds shall transmit imposed stresses without permanent deformation of failure when subjected to proof or service loading.

3.8.5 Enclosure. The distribution panel enclosure shall be designated to provide adequate weather protection under all environmental conditions expected to be encountered. The enclosure shall be provided with adequate ventilation to afford satisfactory performance in ambient temperatures of 125 degrees Fahrenheit (^{o}F) 52 degrees Celsius (^{o}C). Drain holes shall be provided in the lowest part of the enclosure to drain off any moisture that may accumulate inside the enclosure. Carrying handles in accordance with MS 18012 shall be provided when specified on the applicable specification sheet.

3.8.5.1 Cover. A hinged heavy-duty cover shall be provided for access to the area of the handles of all circuit breakers. Positive latching hand-operated catches shall be provided to hold the cover in a closed position. The cover shall positively latched at a position of 20 to 30 degrees past vertical when in the full open position and shall be manually closed. The cover shall be designed to provide a weatherproof seal when closed. The cover shall be fabricated from heavy-gage aluminum and shall be reinforced, as required, such as that the distribution panels can be stacked without damage three high, one on top of the other, with the weight of each distribution panel distributed on the cover of the distributed panel beneath without the use of any intervening material to redistribute the weight between distribution panels.

3.8.5.2 Hinged access panels. Unless otherwise specified in the applicable specification sheet, top hinged access panels with positive latches shall be provided in a manner to permit ready access to the interior of the distribution panels for making power connections, adjustments, and changing transformer tap positions, repairs, or replacement of parts.

3.8.5.2.1 Paralleling cable access panel. When specified in the applicable specification sheet, one paralleling cable entrance access panel shall be provided for paralleling two distribution panels. This access panel shall be similar to the power input entrance panel and shall be located on the distribution panel enclosure adjacent to the paralleling terminals in a manner that provides ready access to the terminals. The access panel shall be hinged at the top.

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3.8.6 Bus bars. Bus bars shall have sufficient capacity to provide the load rating as specified on the applicable specification sheet. The bus bars design shall meet the requirements of UL 67 for the specification sheet bus rating. Bus bars shall be of a copper or aluminum material. Aluminum bus joints and connector surfaces shall be plated in accordance with UL 67.

3.8.7 Wire terminals. All wire connectors shall conform to UL 486A. Circuit breaker terminals shall not be used as input or output terminals. The terminal size and capacity shall be adequate to support the wire size required for the related circuit loads. Pressure type connectors shall be of heavy-duty construction and provide means for securing the bare connector. Connectors shall be tin plated. When allen screws are used as the pressure applying means, one hex or allen wrench of each size required shall be furnished when specified (see 6.2). Means shall be provided to securely store the hex wrenches on the inside of the distribution panel in an easily accessible location. All input, output, feedthrough, paralleling, and branch circuit pressure type terminal connections shall be connected to heavy-duty terminal blocks.

3.8.7.1 Terminal access. Cable entry to input, output, and input paralleling terminals shall be through one or more holes located on the bottom of the distribution panel. The holes shall have insulating bushings to prevent chaffing of the input power cables on the hole edges. Access to terminals shall be by means of hinged access panels (see 3.8.5.2).

3.8.7.2 Input terminals. When input terminals are specified in the applicable specification sheet, the input terminals shall be of copper or aluminum material. If aluminum, the terminals shall be plated in accordance with UL 67 for use with copper connectors. For 3 phase input power, the terminals shall be permanently and legibly marked T1, T2, T3, T0 and G corresponding to phase A, phase B, phase C, the neutral (N) and ground (G) respectively.

3.8.7.3 Output or feedthrough terminals. When output or feedthrough terminals are specified in the applicable specification sheet they shall be located in the proximity of the input terminals. The output terminals shall be used for extending electric power to other remotely located distribution panels. The output terminals shall be of the same type, size and material as the input terminals (see 3.8.7.2). For 3 phase input power, the output terminals shall be permanently and legibly marked L1, L2, L3, L0 and G corresponding to phase A, phase B, phase C, N, and G respectively.

3.8.7.4 Paralleling terminals Paralleling terminals shall be provided when specified in the applicable specification sheet. These paralleling terminals shall be used to increase the power distribution capability in the immediate area. The paralleling terminal connectors shall be pressure type connectors of a type and size to accept the current carrying conductor of single conductor, copper cable. The terminals shall be permanently marked P1, P2, P3, and G corresponding to phase A, phase B, phase C, N and G respectively.

3.8.8 Grounding. The distribution panel enclosure, frame, all non current carrying metal parts, grounding terminals and grounding connector shall all be electrically bonded together. Grounding shall conform to UL 67 and NFPA 70.

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3.8.8.1 Grounding connector. A solderless, pressure type, ground connector shall be provided on all distribution panels. This ground connector shall be conveniently located and of such size for connecting a ground conductor conforming to NFPA No. 70 and UL 67. The ground connector shall extend through the frame of the distribution panel enclosure to the exterior of the enclosure.

3.8.9 Transformer. Dry type, totally enclosed, two winding (high and low voltage) transformers with four 2-1/2 percent full capacity taps (two above and two below the nominal input voltage) shall be furnished and installed in the type 2 distribution panels. The transformers shall be furnished with suitable tap changing provisions. The dry type transformers shall have an encapsulated 185°C insulation system rated for 115°C rise. The transformers shall be in accordance with NEMA ST 20 and UL 506.

3.8.9.1 Transformer cooling. Ventilation and heat dissipation means shall be provided for removing the heat generated by the transformer at rated load from the interior of the distribution panel to meet the requirements of NEMA ST 20 and UL 506.

3.8.9.2 Transformer noise. For type II distribution panels containing a single transformer, the average sound level generated by each transformer shall not exceed 34 decibels (db) for transformers rated 0-9 kilovolt-amperes (kVA) and shall not exceed 42 dB for transformers rated 10-50 kVA when measured as specified in NEMA ST 20. For distribution panels containing three identically rated transformers, the average combined sound level of all three transformer shall not exceed 40 dB for transformers rated 0-9 kVA and shall not exceed 48 dB for transformers rated 10-50 kVA when measured as specified in NEMA ST 20.

3.8.9.3 Transformer installation. Each transformer shall be installed in such a manner that it is rigidly fastened to, and its weight supported by, the frame. The sheet metal bottom or sides of the distribution panel enclosure shall not be used to provide transformer support.

3.8.10 Panelboards. Panelboards shall conform to UL 67.

3.8.10.1 Ventilation. The design of the distribution panel shall provide for air circulation through the interior of the distribution panel.

3.8.10.2 Sound level. The sound level of type II distribution panels shall not exceed the limits specified in 3.8.9.2 for transformer generated noise. Only type II panels need be treated for sound levels.

3.8.10.3 Power-on indicator. Power-on indicators shall be provided on the circuit breaker panel to indicate when the main buses are energized. The indicator shall illuminate red light when the bus is energized. For 3 phase power, an indicator shall be provided for each phase.

3.8.10.4 Wiring identification. All wire conductors within the distribution panel shall be marked for identification. The identification shall include a circuit number, if applicable, and the function, including phase A, phase B, phase C, N or G. Conductors connected between the terminals

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of connectors and the circuit breakers shall be further identified by the color code of the conductor as specified in the applicable specification sheet. Color code bands may be used on the conductor ends in lieu of colored conductors.

3.8.11 Circuits. All single phase circuits shall be distributed between the three phases in such a manner to maintain load balance as close as possible.

3.8.11.1 Circuit breaker. The circuit breakers shall be of the molded case, thermal-magnetic type having the number of poles, current rating, and voltage rating as specified in the applicable specification sheet. The circuit breakers shall conform to W-C-375 and UL 489. Circuit breaker interrupt capacity shall be 14,000 amperes symmetrical minimum.

3.8.11.1.1 Ground fault circuit interrupter (GFCI) breakers. All single pole, 15 ampere and 20 ampere rated, 120 V, 60 Hz, single phase circuit breakers shall be the GFCI type and shall conform to UL 943.

3.8.11.2 Circuit identification. The circuit breakers and corresponding circuit receptacles or interface connectors shall be identified by the same permanently affixed and permanently legible identification scheme. The identification shall include the circuit number, current rating, circuit voltage, and phase identification. Phase identification shall be single phase, or 3 phase, as applicable, and the actual phase(s) connected to the circuit, that is, for single phase: phase A, B, C, AB, BC, or CA. The identification shall be placed on the visible side of the circuit breaker panel adjacent to the circuit receptacle or interface connector.

3.8.12 Connectors. All connectors shall conform to the requirements specified in the applicable specification sheet. For receptacle and plug type connectors, the connector and its shell housing shall be capable of forming a watertight joint at the junction where the mating plug or cover is attached to the connector and secured in place. Connectors shall be located on the lower, inward sloping face portion of the sides of the distribution panel. The configuration of the connectors shall be such that the quantity, kind, and arrangement of connectors on one side shall be the same, or as nearly the same as possible, as the connector configuration on the other side. The connectors shall be arranged so that the distribution panel will remain in a balanced upright position when being lifted, in order to comply with the requirements of 3.9. For receptacle and plug type connectors, waterproofing gaskets shall be used between the connector and the distribution panel.

3.8.12.1 Connector terminal wiring. NEMA type connectors and military type connectors conforming to MIL-C-22992, class L type only, shall be wired in accordance with one applicable specification sheet wire termination table.

3.8.12.2 NEMA type connectors. NEMA type connectors shall be locking type single connectors. The connectors shall conform to NEMA WD 1, NEMA WD 6, W-C-596, and as specified in the applicable specification sheet, and shall meet the applicable test provisions of 4.4. The connector bodies shall be made of a heavy duty, physical shock resistant, temperature resistant, sunlight resistant,

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molded plastic compound having a molded-in color. A mounting shell with a hinged, gasketed, watertight cover shall be provided for mounting the connector body on the distribution panel. The cover shall be made of fiberglass or polycarbonate plastic material with stainless steel or plated brass hinge hardware and mounting screws.

3.8.12.3 Military type connectors. Military type connectors shall conform to MIL-C-22992.

3.8.12.4 Pressure type terminal connectors. Pressure type connectors shall be mounted on terminal blocks and arranged in groups with the terminals within each group arranged in the order phase A, phase B, phase C, N and G.

3.8.13 Interior wiring system. All electrical materials used in the distribution panels, except for military type connectors, whether specified herein or not, shall be listed in the UL Electrical Construction Materials List and bear the UL listing mark. The installation of all components into the distribution panels shall be in accordance with the requirements of NFPA No. 70 and UL 67. Each circuit shall be continuous (without splices) including the neutral and equipment ground, when required. All inter-connecting wiring shall be of copper or copper alloy material.

3.8.14 Wiring diagram. A weather resistant, permanently legible, color coded wiring diagram shall be furnished with each distribution panel and shall show all electrical circuits including the identification of all components and terminal assignments and the color identification from the wire from-and-to each terminal point. This wiring diagram shall be permanently attached on the inside of the distribution panels access door covering the circuit breaker trip handles (see 3.8.5.1) and shall be easily readable when the cover is in the fully raised position.

3.9 Lifting attachments. Unless otherwise specified in the applicable specification sheet, suitable lifting attachments shall be provided to enable the distribution panel to be lifted in its normal position and remain within 15° of level when lifted by crane. Each attachment shall be designed to carry a minimum of four times the weight of the distribution panel. When welded construction is used, the weldment shall be reinforced by bolts, each bolt capable of carrying 2-1/2 times the weight of the panel. For distribution panels weighing over 100 pounds, provisions shall be made for lifting the distribution panels by forklifts. The lifting attachments, when not in use, shall not increase the overall height of the panels.

3.9.1 Tiedown device. The distribution panel shall be provided with suitable permanently attached devices to permit tie down of the fully assembled panels to the floor or deck of the transporting medium. The tiedown devices shall be of such design, location, and number that will enable the panel to be anchored without shifting in any direction. The distribution panels shall be capable of being restrained to and withstand, without loss of serviceability, the following forces applied statically and independently: Fore, 3G; aft, 1-1/2G; lateral, 1-1/2G; up, 2G; and down, 4-1/2G. The distribution panels shall also withstand the above forces applied dynamically for 0.1 second (1G is equal to the weight of the fully assembled panel).

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3.10 Fungus treatment. The distribution panels shall be given a type II treatment for fungus resistance in accordance with MIL-T-152.

3.11 Treatment and painting. Treatment, cleaning, and painting selection and processing, shall be selected from the following options depending on the application and environment (see 6.2). Only lead- and chromate-free materials shall be used.

3.11.1 Standard cleaning and painting option. All inside and outside surfaces of the distribution panels, except for the electrical components installed, shall be coated. Surfaces to be coated shall be cleaned and dried to insure that they are free from contaminants, such as oil, grease, welding slag, loose welding splatter, loose mill scale, water, dirt, corrosion products or any other interfering substances. As soon as practicable, after cleaning and before any corrosion product or other coating interfering material can result, the surface shall be prepared or treated to insure the adhesion of the coating system. The coating shall be applied to the clean, dry surface immediately after cleaning and treating. The coating shall be epoxy polyamide, or polyester resin, or equal. Electrostatic coating powder is acceptable. The total dry film thickness shall be not less than 2.5 mils over the entire surface. The coating shall be free from runs, sags, orange peel, or other defects. The color of the finish coat shall be green color #14064 of FED-STD-595.

3.11.2 Chemical agent resistant coating (CARC) option. Treatment of the entire distribution panel shall be in accordance with MIL-T-704 and as stated herein.

a. All external surfaces of the distribution panel, regardless of the material selected, shall have a finish coat of CARC paint in green 383, color #34094 of FED-STD-595.

   (1) Where applicable to the material selected, all exterior surfaces shall be painted in accordance with MIL-T-704, Type F or G.
   (2) Where the exterior material selected is not applicable to MIL-T-704 (use of plastics or composites as an example) the surface shall be treated as necessary for application of the finish coat in accordance with MIL-C-46168.

b. All other distribution panel surfaces, to include those within a panel housing, those behind insulation material, and the insulation retainer, if used, shall be treated using manufacturer's commercial practices and painted in accordance with MIL-T-704 with finish coat of paint in accordance with TT-E-1593. Color shall be green 383, color #34094 of FED-STD-595.
c. The following items shall not be painted: terminal wiring connections, diagrams and plates, relays, switches, circuit interupters, instrumentation, rubber, all other parts whose operation or function would be adversely affected. Insulation material shall be painted unless the sound absorbing characteristics of the material are compromised.
d. When specified, a camouflage pattern, to be supplied by the Contracting Officer, shall be applied to the distribution panel. Application of the pattern shall be in accordance with MIL-T-704.

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e. All exterior surface labels and markings specified in this document shall be applied on camouflage painted panels using the following coloration scheme:

(1) Equipment that is painted with a standard 3-color camouflage pattern, the following letter colors shall be used:

BACKGROUND COLOR	LETTER COLOR
Green 383	Black
Brown 383	Black
Black	Green 383

(2) Equipment that is camouflage painted (temporarily or permanently) for winter conditions, the following letter colors apply:

BACKGROUND COLOR	LETTER COLOR
White	Black
Brown 383	Black
Black	Green 383

(3) Equipment that is camouflage painted (temporarily or permanently) for desert conditions, the following letter colors apply:

BACKGROUND COLOR	LETTER COLOR
Tan 686	Brown 383
Brown 383	Black
Black	Brown 383

f. All specified identification plates applied to exterior surfaces shall be installed following the final finish coat. Should a camouflage pattern be specified, placement of the plates should be on the black camouflage pattern area when possible.

g. CARC paint shall not be applied to surfaces that will exceed temperatures of 400°F (204°C).

3.12 Environmental requirements. The distribution panel shall be capable (see 6.2) of satisfactory operation under the following environmental conditions specified herein. Unless otherwise specified, certification in lieu of tests is acceptable, where noted.

3.12.1 Low temperature. The distribution panels shall operate satisfactorily in an ambient temperature of -25°F (-32°C) as tested in accordance with 4.4.4.1. Certification is acceptable.

3.12.2 High temperature. The distribution panel shall operate satisfactorily in an ambient temperature of 125°F (51.7°C) as tested in accordance with 4.4.4.2.

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3.12.3 Rain. The distribution panels shall operate satisfactorily in windblown rain of 2 to 5 inches per hour as tested in accordance with 4.4.4.3.

3.12.4 Salt fog. The distribution panels shall be capable of being stored and operated in a salt fog atmosphere without effect in performance as tested in accordance with 4.4.4.4. Certification is acceptable.

3.12.5 Sand and dust. The distribution panels shall operate satisfactorily in a sandy and dusty environment as tested in accordance with 4.4.4.5. Certification is acceptable.

3.12.6 Shock. The distribution panels shall operate satisfactorily after being subjected to impact shocks, such as handling and transportation shocks, as tested in accordance with 4.4.4.6.

3.12.7 Humidity. The distribution panels shall operate satisfactorily in a humid atmosphere of 95 ±5 % relative humidity as tested in accordance with 4.4.4.7. Certification is acceptable.

3.13 Instruction and warning markings. All marking on plates and labels shall be permanent. Markings shall be etched, inscribed, or stamped.

3.13.1 Instruction plates. The distribution panel shall be equipped with instruction plates suitably located, which describe any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts, or rivets of not less than 1/8-inch diameter. A plate marked "ground" shall be installed at the grounding terminal (see 3.8.8) on the exterior of all distribution panels.

3.13.2 Warning labels. All access panels (see 3.8.5.2) and the circuit breaker panel shall have a permanently legible warning label permanently affixed to the outside of the access panel, indicating the danger of exposed energized electrical connections on the inside of the distribution panel. Warning labels shall be made of corrosion-resistant aluminum or brass and shall be affixed to the access panel by nonferrous metal screws or rivets.

3.14 Identification plate. A nameplate of corrosion-resistant metal shall be installed on the face of the circuit breaker panelboard with nonferrous metal screws, rivets, or bolts of not less than 1/8-inch diameter. The nameplate shall be permanently and legibly marked by inscribing or stamping manufacturer's model number and serial number, power handling capacity in kilowatts, voltage, phase, frequency and current rating of input, dimensions, weight, date of manufacture, contract number, national stock number, and military specification reference.

3.15 Certification. When the option to certify to the environmental requirements specified herein is exercised, the supplier shall provide a certified statement that the panels meet the requirements as specified and provide supporting test data when requested by the acquisition activity..

3.16 Workmanship.

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3.16.1 Metal fabrication. The metal used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the metal to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.16.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be properly tightened.

3.16.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.16.4 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

3.16.5 Electrical. Circuit breakers shall be securely fastened to the panel board buses. Wire and cables to the circuit breakers shall be securely fastened to the circuit breaker terminals. Only the minimum amount of insulation shall be removed from the wires or cables for electrical connections to the circuit breakers, connectors, terminals, or other devices. Termination of connectors in terminals and connectors may be by pressure or soldering processes. All soldering materials and workmanship shall meet the requirements of DOD-STD-2000-4. Wires and cables shall be of adequate length and neatly routed inside the distribution panel near the sides or bottom of the distribution panel. Care should be taken to see that no strain is placed on circuit breaker terminals or connector terminals. Cable shall be secured by clamps or standard ties as applicable.

3.17 Manuals. Operational maintenance manuals required for proper operation and servicing shall be provided with the equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Standards compliance. The contractor shall make available to the contracting officer or his authorized representative evidence of compliance with

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the applicable standard(s) cited in 3.6.1. The Government reserves the right to examine and test all delivered panels to determine the validity of the certification.

4.1.2 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.3 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one distribution panel of each type, class, configuration when a first article is required (see 3.3 and 6.2). This inspection shall include the examination of 4.3 and the tests of 4.4. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.4.1 and 4.4.2.2, and the packaging inspection of 4.5. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Examination. Each distribution panel shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4 Tests. Failure of the distribution panels tested to meet the requirements of the tests specified herein shall constitute cause for

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rejection. Unless otherwise indicated, all circuits shall be energized and delivering power to loads during the tests, except for the dielectric strength test (see 4.4.3), shock test (see 4.4.4.6), stability test (see 4.4.5), and tiedown test (see 4.4.6). When a first article inspection is specified (see 4.2.1), the tests of 4.4.2 and 4.4.3 shall be performed prior to and following environmental tests of 4.4.4.

4.4.1 Electrical design test. Each distribution panel shall be checked both visually and electrically for conformance to the electrical design. The distribution panel shall be energized for electrical voltage checks only. Wiring accuracy and continuity shall be verified. The test shall encompass all aspects of the distribution panel electrical design, including components, proper circuit identification, proper terminal and connector wiring, transformer and transformer tap changing wiring, GFCI, circuit breaker wiring, secureness of wiring connections, grounding, manual electrical disconnect operation of direct breakers, proper functioning of indicators, conformance to NFPA 70, and overall electrical design integrity. Suitable electrical test instruments shall be used to verify proper design operation. Voltages shall be verified at all connector terminals. On type II distribution panels, applicable tap changing provision shall be checked for proper operation in all tap positions, and transformer voltages shall be verified with design for all tap positions.

4.4.2 Operational tests.

4.4.2.1 Circuit breaker tests. Circuit breaker circuits shall be loaded to 100 percent of their rated current loading for 1 hour. After being under this load for 1 hour, each circuit shall be separately tested for overload response by imposing a load of 200 percent current rating of the applicable breaker. All breaker poles shall be loaded equally. Each circuit breaker shall open under the imposed load within the time limits established in UL 489. At the conclusion of the tests, any evidence of abnormal circuit breaker operation, or evidence of wire damage shall be cause for rejection.

4.4.2.2 GFCI tests. Each GFCI type breaker will be tested for proper GFCI operation by following the breaker manufacturer's recommended test procedure using the built-in test circuitry of the GFCI breaker.

4.4.3 Dielectric strength test. The panel shall be tested for dielectric strength in accordance with UL 67.

4.4.4 Environmental tests. The distribution panel shall operate satisfactorily under the branch breaker loads and conditions specified in 4.4.4.1 through 4.4.4.7. Satisfactory operation shall constitute no inadvertent breaker operation at rated current loads, component failures, or evidence of impending failure.

4.4.4.1 Low temperature test. The low temperature test shall be conducted in accordance with MIL-STD-810, method 502.3, procedure I at a minimum temperature of -25°F (-32°C) and a duration, after panel stabilization, of 24 hours. The distribution panels shall operate satisfactorily at the peak of the most severe level and after return to room ambient conditions at the rated circuit breaker current loading.

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4.4.4.2 High temperature test. The high temperature test shall be conducted in accordance with MIL-STD-810, method 501.3, procedure II, operation, constant temperature exposure. The panel shall be maintained at a stabilized temperature of 125°F (52°C) for a period of 48 hours. The distribution panel shall operate satisfactorily during the high temperature exposure and after returning to room ambient at rated current loading.

4.4.4.3 Rain test. The test unit shall be rain tested by the following procedure:

a. Place test unit in a test enclosure in the normal operating position.
b. Produce a simulated rainfall at a constant rate of 2 ±0.5 inches per hour.
c. Rainfall shall cover a total arc of 90°, 45° on either side of vertical, for a total time not less than one hour.
d. Rotate test unit 90° about vertical axis from its position in step a and repeat steps b and c.
e. The inability of the test unit to operate satisfactorily at rated loads during and after the rain test shall constitute a failure.

4.4.4.4 Salt fog test. The salt fog test shall be conducted on the distribution panel in accordance with MIL-STD-810, method 509.3 procedure I.

4.4.4.5 Dust test. The dust test shall be conducted on the distribution panel in accordance with MIL-STD-810, method 510.3 using an air velocity of 1750 feet per minute.

4.4.4.6 Shock test. The test unit shall be shock tested by the following procedure:

a. Place test unit in a normal operating position.
b. Raise test unit six inches vertically and free drop on a two inch thick plywood surface backed by concrete.
c. Repeat step b three times.
d. Upon completion of the drop test, the test unit shall be visually inspected for damage and operation.
e. The inability of the test unit to operate satisfactorily at rated loads or indication of structural damage shall constitute a failure.

4.4.4.7 Humidity test. Prior to conducting these tests, the condensed moisture shall be removed from the exterior of the test panel(s), and the panel(s) shall be effectively grounded, with the ground resistance not exceeding 25 ohms in accordance with NFPA 70. The humidity test shall be conducted in accordance with MIL-STD-810, method 507.3, procedure I, cycle I, except that the maximum temperature shall be 125°F (52°C) and the total number of cycles 5. After the conclusion of the fifth cycle the panels shall operate satisfactorily at rated circuit breaker current loads.

4.4.5 Stability test. Dummy cable and connector assemblies having the same weight, moment distribution, and mounting orientation as the actual cable and connector assemblies shall be attached to all the mating receptacles or lug terminals on one side of the distribution panel. The distribution panel

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shall then be placed on a 15° incline, with the panel side having the connector cables attached, facing down the incline slope. The distribution panel shall be stable and show no signs of tipping when the cables are lying on the incline in a stable position. Disconnect the connector assemblies and install the connector assemblies on another side of the distribution panel. Repeat the test until each side having connectors has been tested.

4.4.6 Tiedown test. The distribution panel shall have the forces specified in 3.9.1 applied. No evidence of damage to or bending of the tiedowns or distribution panel frame and housing shall be evidenced.

4.4.7 Stacking test. Three distribution panels of the same type, class, and configuration shall be stacked, one placed on top of the other, without the use of any intervening load distributing material. The distribution panels and covers shall show no evidence of damage as specified in 3.8.5.1.

4.4.8 Type II and sound level test. The distribution panelboard sound level for type II only shall be tested as specified in NEMA ST 20, except that the distribution panel shall be considered as the noise source instead of the transformer, and, where applicable, all measurements shall be made relative to the distribution panel instead of the transformer. The sound level shall conform to 3.8.10.2.

4.5 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-E-17555. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and packing, and marking shall be in accordance with the requirements of section 4 of MIL-E-17555 (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class, configuration of panel required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).
d. When a first article is required for inspection and approval (see 3.3, 4.2.1 and 6.3).

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e. Title, number, and date of the applicable specification sheet (see 3.1).
f. When hex wrenches are required to be furnished (see 3.8.7).
g. Treatment and paint option (see 3.11).
h. If certification in lieu of test is not applicable (see 3.12).
i. When a first article pack inspection is required and time frame required for submission (see 4.2.1 and 4.5).
j. Level of preservation and level of packing required (see 5.1).

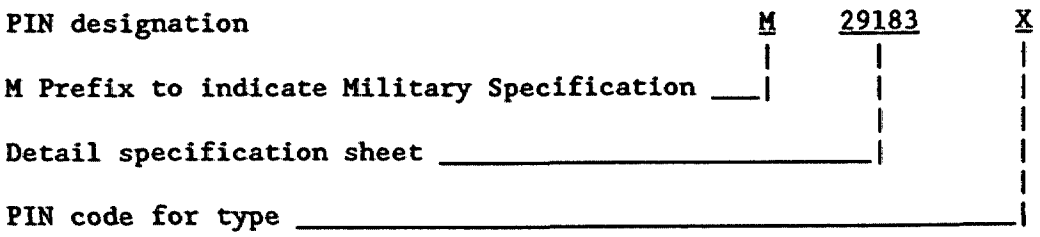
6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of number of units required by the using agency contracting officer. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Metric equivalents. Metric equivalents, indicated herein and in the specification sheets MIL-P-29183/XX are for information only. In each instance the value stated in U.S. customary units shall be controlling.

6.5 Subject term (key word) listing.

Compliance
Cover
Bus
Terminals
Transformer
Wiring identification
Circuit breakers
Connectors
Interior wiring
Environmental test
Design test
Data
First article
Part identification

6.6 Part or identifying Number (PIN). The Government PIN should be structured as follows:



6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Review activities:
Navy - MC, MS
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DLA - GS

Preparing activity:
Navy - YD

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-P-29183B MIL-P-29183/04,05,15 to 20	2. DOCUMENT DATE (YYMMDD) 1990 June 18

3. DOCUMENT TITLE GENERAL SPECIFICATION FOR PANELBOARDS, POWER DISTRIBUTION, PORTABLE, WEATHERPROOF

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME JOHN E. SCHROYER, Director Facilities Criteria Development Div	b. TELEPHONE (Include Area Code) (1) Commercial (805) 982-5801 x402	(2) AUTOVON 551-5801 x402
c. ADDRESS (Include Zip Code) Commanding Officer (156) Naval Construction Battalion Center Port Hueneme, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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